DPOE Autoresearch Stress Test

Closed experiments, bounded claims, proposed plan iteration

Agent-prepared factual material; critique and reflection reserved for the human

Lossfunk Autoresearch Challenge

Starting research question and claims

Question: Can calibrated MLLM reward uncertainty improve visual-domain adaptation when the same epistemic signal is used pessimistically and optimistically?

1. C1: calibration matters more than reward-model capacity.
2. C2: epistemic, not aleatoric or matched noise, drives benefit.
3. C3: coupled reward pessimism (P), adaptive KL (K), and optimistic collection (O) beats either half.

The submitted plan targeted Qwen2.5-VL models, visual manipulation, 5 seeds, and millions of environment steps. The sprint tested only CPU-scale mechanisms.

Autoresearch flow

Driver: OpenAI Codex

- hardware/literature audit
- MVE design and implementation
- locked execution and analysis
- iterated plan, paper, factual deck

Reviewer: Claude Code

- independent checkpoint reviews
- blocking validity corrections
- some manual human-mediated captures after CLI failures
- human account; outside the $50 sprint budget

Human authority controlled session time, experiment sign-off, scope changes, manual review delivery, and experiment closure.

How Voila was customized

The generic exploration workflow was replaced by a fixed-plan stress test:

- no new topic selection: the screened DPOE plan was binding;
- hardware audit before MVE design; full MLLM plan explicitly out of scope;
- 2–3 falsification-oriented MVEs with support/kill criteria;
- Claude review and human sign-off before execution;
- `PROGRESS.md`, `LEARNINGS.md`, and `BUDGET.md` as resumable audit trails;
- negative/null/untested results treated as first-class outputs;
- human-only critique and reflection protected from agent drafting.

Sessions were time-boxed by the human (3h, 8h, then 4h blocks).

Compute-driven de-scope

Available	Consequence
4 CPU cores / 8 threads	NumPy-scale synthetic tests
5.8 GiB RAM + 4 GiB swap	no local multi-billion-parameter MLLM
No accessible NVIDIA GPU	C1 MLLM comparison deferred
$50 ceiling; about $20 driver inference	local experiments cost $0

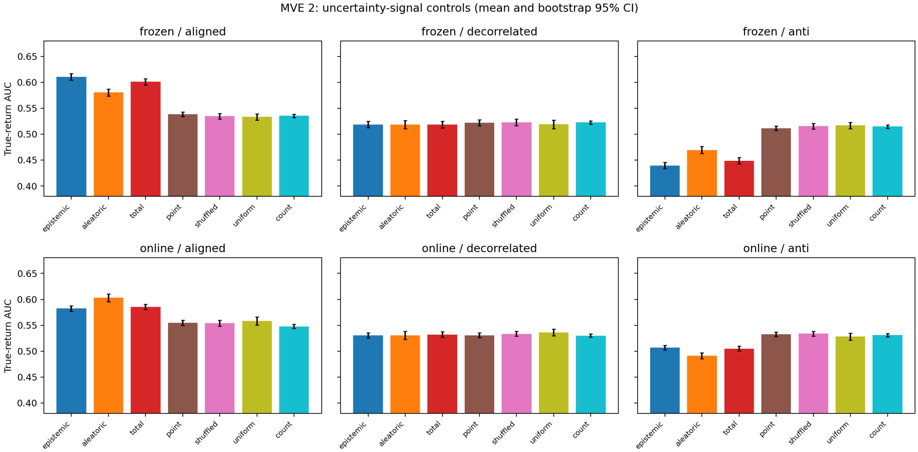
MVE 1 never ran. MVE 2 tested the frozen/online uncertainty mechanism; MVE 3 tested coupling validity and solvability.

MVE 2: frozen uncertainty did not acquire information

- 50 locked seeds; three shifts; aligned, decorrelated, and anti-correlated layouts.
- Frozen epistemic minus point: $+0.0719$ aligned, -0.0036 decorrelated, -0.0719 anti.
- Visits cannot update a frozen reward model: apparent gain followed static alignment.
- Online epistemic beat aleatoric by $+0.0081$ [0.0030, 0.0131], but the estimator was weak (AUROC 0.532; Spearman 0.162) and generic controls did better.

Verdict: the frozen information-gain rationale is structurally killed; the broader strong-online-estimator claim is unresolved.

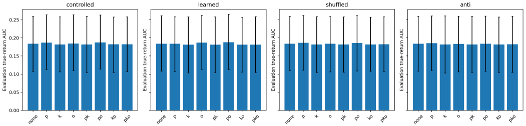
MVE 2 result landscape



MVE 3 v1: the apparent kill was invalid

- 4,800 locked factorial cells; controlled signal passed quality and leakage gates.
- Preregistered rule recorded *killed-at-toy-scale*.
- Independent review exposed exploitation true-goal AUC 0.0004 (P+K+O: 0.0002).
- Kill withdrawn via disclosed post-review validity amendment.

Final v1 status: untested-design-floor-policy-collapse.



MVE 3 v2: bounded solvability failure

Complete preregistered 27-tuple development grid:

- best tuple: $\tau_{ref} = 0.18$, $\beta_0 = 0.10$, clip ± 1.75;
- oracle goal success 16.44% vs required 25%;
- oracle-minus-none gap 11.36 pp vs required 15 pp;
- return-AUC separation passed at 0.3485;
- best tuple was on the relaxation boundary.

Conclusion: not solvable within the preregistered grid.
No adaptation-engagement gate, pilot-v2, or locked-v2 seed was opened.

Closed claim table

Claim	Status
C1	**Untested.** MVE 1 did not run; DriveReward forces specialization/data/reward-scale controls in any future test.
C2	**Frozen information-gain rationale structurally killed analytically and corroborated by MVE 2.** Online claim unresolved.
C3	**Untested at toy scale.** v1 exploitation floor; v2 not solvable within the pre-registered grid.

Proposed research-plan delta

1. Separate frozen-RM risk control from online-RM active learning.
2. Gate estimator claims on shifted error prediction and calibration.
3. Require oracle reachability, return headroom, and factor-independent adaptation engagement before a coupling factorial.
4. Use a complete $P \times K \times O$ factorial with practical effect floors and explicit hacking denominators.
5. Reframe C1 around calibration *after* controlling specialization, data, reward scale, calibrated-large, and matched compute.

This is an agent-proposed revision; the human decides the final delta.

HUMAN-ONLY: Merciless critique of the AI artifact

PLACEHOLDER — HUMAN AUTHOR ONLY

The agent intentionally supplied no critique content on this slide.

HUMAN-ONLY: Reflection on autoresearch limits

PLACEHOLDER — HUMAN AUTHOR ONLY

The agent intentionally supplied no reflection content on this slide.

Appendix: exact prompts and user decisions

Reviewer invocation pattern:

```
claude -p "REVIEW REQUEST: <what you did and why>.  Files to review:  <paths>."
```

Human decisions retained verbatim:

- *Approved, start MVE 2 now, log the sign-off.*
- *Final sign-off: implement and execute MVE 3 exactly under the approved protocol. Ensure the locked run can resume from the last completed cell after an interruption rather than restarting. Log this as a user decision.*
- *Experiments are closed. MVE 1 will not run.*

The full session narrative and additional decisions are in `PROGRESS.md`.

Appendix: reviewer trail highlights

- Reviews 01–07: MVE protocols revised to approval before locked execution.
- Review 08: exposed v1 exploitation floor; prevented an invalid C3 kill.
- Reviews 09–14: forced deviation disclosure, solvability gates, audit repair, and report-only regeneration.
- Review 15 was REVISE, not the initially reported APPROVE; stale v1 pilot naming fixed.
- Review 16: restored related work, reviewer provenance, missing deviations, accurate title, and compilable paper.
- Review 17: corrected the inherited Zhang/AdvPO citation to the stored Kang–Oh ICLR 2025 paper.

Some later reviews were human-mediated manual Claude runs after empty/API-error CLI captures.

Appendix: time and cost actuals

Work	Actual	Dollar cost
MVE 2 locked run	257.7 s	$0
MVE 3 v1 locked run	8,462.3 s (2.35 h)	$0
MVE 3 v2 report-only audit	same development namespace	$0
Driver inference	session-spanning	about $20

Claude review used the human's personal account and was outside the $50 sprint budget.

Appendix: artifact map

- Paper: `paper/dpoe_negative_result.pdf`
- Proposed plan: `iterated-research-plan.md`
- MVE 2: `mve2/`, `results/mve2-full/`
- MVE 3 v1: `mve3/`, `results/mve3-locked/`
- MVE 3 v2 gate: `mve3v2/`, `results/mve3v2-development/`
- Audit trail: `PROGRESS.md`, `LEARNINGS.md`, `BUDGET.md`, `reviews/`